Hassan Shahzad

Software Engineer

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Summary

Passionate full-stack and AI software engineer with over 3 years of experience building scalable web applications and intelligent systems. I specialize in modern frameworks, cloud technologies, and intelligent solutions. My work focuses on designing end-to-end systems that strike a balance between robust backend architecture and intuitive user-facing experiences. Built platforms that handle real-time interactions, scalable data flows, and intelligent decision-making powered by advanced AI techniques. Beyond core engineering, I enjoy experimenting with emerging tools and translating complex ideas into practical, impactful products.

Experience

Software Development Engineer - I

Arrivy | *02/2025 – 06/2025* (5 mo)

Arrivy is a field service coordination platform that streamlines customer communication, crew management, and operational logistics in real time.

- Built and maintained Python and Node microservices for PDF handling and database tasks, improving modularity and deployment speed.
- Implemented dynamic PDF generation and compression pipelines using Puppeteer and Ghostscript, reducing PDF sizes by 80% without loss in quality.
- Developed and deployed Google Cloud Functions and Pub/Sub pipelines to enable real-time crew attendance tracking, logging schedules, and work hours with high reliability.
- Leveraged Redis queues and deferred services to run background tasks asynchronously, cutting user-facing latency by over 60% during PDF processing.
- Built an internal authorization tool (React + Flask) for account access management, improving auditing and traceability.
- Designed and implemented a React-based template system for task creation, resulting in files attached to templates auto-copied to tasks on selection, accelerating workflows and reducing manual effort.
- Implemented secure DB authentication using sandboxed service accounts to access production data via SQLAlchemy ORM.
- Integrated third-party APIs (HubSpot, Workday, Easybook) for real-time data sync and workflow automation.

Technical Content Engineer

Educative | *07/2023 – 01/2025* (1 yr 6 mo)

Educative is an interactive learning platform for software developers to master coding, system design, and technical interviews through hands-on courses.

- Curated technical content and real-world projects for learners focused on React and Flask, boosting course completion rates through hands-on learning.
- Designed and maintained backend-centric projects on Flask, databases, and REST APIs, covering auth flows, DB operations, and production deployment.
- Developed custom Docker images to run complex coding environments directly in-browser, reducing setup friction and support requests.
- Built real-world coding challenges aligned with interview patterns, boosting learner retention and job-readiness scores.
- Integrated Generative AI into coding widgets for real-time mentorship and feedback using OpenAI APIs, increasing learner engagement by 25%.
- Contributed to the company's AI mock interviewer by leading prompt engineering, testing scenarios in playgrounds, and refining feedback quality.
- Collaborated with engineering and content teams to deliver reliable, browser-ready environments for thousands of concurrent users.

Engineering Intern

Contract.pk | *07/2022 – 09/2022* (3 mo)

Contract.pk is a digital platform that simplifies contract creation, signing, and management for businesses and individuals in Pakistan.

- Built a legal services platform using Flask for backend APIs and React + Tailwind for a responsive and modern UI.
- Designed RESTful API endpoints for secure case data handling and streamlined client-lawyer communication.
- Developed intuitive frontend components using TailwindCSS, improving user navigation and accessibility.
- Integrated Flask and React to enable dynamic case workflows, reducing manual steps and increasing active user time by 20%.

Education

FAST National University of Computer and Emerging Sciences

B.S. Computer Science | *08/2019 – 06/2023*

Relevant courses: Web Programming, Software Engineering, Artificial Intelligence, Natural Language Processing, Software Design and Analysis, Software Testing.

Skills

Web Development: JavaScript, React.js, HTML, Tailwind, State Management, Component Lifecycle and Hooks, Callbacks and Promises, Context API, Routing, Custom Components and Reusability, Asynchronous functions

Backend: Flask, Node.js, RESTful APIs, Fast API, Kafka, Sockets, JWT, Cookie and Token Auth, Routing and Blueprints, Request Handling and Validation, Error Handling and Logging, Authentication and Authorization, Pagination and Filtering, Messaging and Queueing, Background Tasks

Databases & Caching: PostgreSQL, MySQL, Redis, SQLAlchemy

AI/ML: Large Language Models, OpenAI API, Natural Language Processing, Prompt Engineering, Transformers, Spacy, NLTK

Cloud & Infrastructure: Google Cloud Platform, Cloud Run, Cloud Functions, Firestore, Cloud Storage, Pub/Sub, Cloud Scheduler, Artifacts Registry, BigQuery, IAM and Role-Based Access Control, Monitoring and Logging

DevOps: Docker, Microservices, Git

Projects

Educative's AI Mock Interviewer

An AI-powered interactive tool designed to simulate real-world technical interviews, providing instant feedback and coding assessments to learners preparing for job interviews.

- Designed and fine-tuned **prompt engineering** workflows to ensure accurate, context-aware interview questions and feedback.
- Tuned **OpenAI API** settings (temperature, frequency penalty, system prompts) for consistent output.
- Helped integrate the AI layer into coding playgrounds, boosting learner engagement and interactivity.
- Integrated AI with browser-based code environments, boosting learner engagement by **25%**.
- Collaborated with product and content teams to align AI-driven assessments with industry standards and real interview expectations.

Wordley.ai

Wordley.ai is an AI tool that analyzes text using NLP and open-source LLMs, providing insights on tone, sentiment, emotion, summarization, classification, and entities.

- Built **transformer** pipelines combining rule-based NLP and open-source models (like **Facebook Bart**, **Twitter Roberta**).
- Integrated **Hugging Face Inference API** for on-demand model execution and scalability.
- Leveraged **NLTK** for lexical and syntactic preprocessing, enabling cleaner input pipelines for transformer models.
- Applied **Textstat** for readability scoring and linguistic complexity insights.
- Developed frontend and backend workflows to visualize structured insight from raw text.
- Gained over **120** users in the first month, with **70%** returning to use the tool again.

SafeLinks

SafeLinks is a microservice platform for creating and managing temporary access links, enabling secure, time-limited sharing of files, URLs, or exclusive pages with automatic expiration and easy integration.

- **Flask** + **Redis** backend handles secure link creation, expiry, and single-use or time-limited access.
- **React** frontend provides real-time link management and status updates for users.
- Dockerized microservices deployed on **Google Cloud Run**, ensuring scalability and isolated service management.
- Integrated Firestore for persistent storage of link metadata and usage logs.
- Implemented secure token-based link resolution to prevent unauthorized access.

Recaller

A smart knowledge base that lets users upload documents and interact with them through contextual search and AI-powered Q&A. It uses embeddings, vector search, and language models to deliver precise answers from uploaded content.

- Built a smart knowledge base enabling users to upload documents (PDFs, DOCX, TXT) and query them using natural language.
- Implemented **vector embeddings** and **similarity search** for contextual retrieval, allowing accurate answers from large datasets.
- Integrated open-source LLMs for AI-powered Q&A, summarization, and contextual responses directly from uploaded documents.
- Designed a **React** + **Flask** stack, managing user uploads, conversation history, and scalable API endpoints.
- Optimized query performance and relevance by combining embeddings with metadata filters and efficient indexing.

Respoon

Respoon is a community-driven platform for sharing surplus food, connecting donors with recipients in real time. It uses maps, location-based search, and real-time chat to make food sharing seamless and efficient.

- Implemented real-time chat with **Flask-SocketIO** for donor-recipient communication.
- Integrated **maps** with geolocation to display nearby food items.
- Added **pagination** and **filtering** for efficient browsing of large datasets.
- Designed robust backend APIs with **Flask** and **PostgreSQL** for managing food items and requests.
- Built an interactive **React** frontend with dynamic filtering, food status updates, and a user-friendly UI.